

Question 8

Q: Given a noisy image acquired from a sensor, explain how the acquisition process contributes to noise and how it can be minimized.

Theoretical Concept: Levels of Computer Vision

During image acquisition, noise is introduced by thermal fluctuations (dark current noise) and electronic circuitry (read noise). This is modeled as additive Gaussian noise or Salt-and-Pepper noise.

Minimization: Can be minimized via hardware (cooling the sensor) or software using Spatial Filters like the **Median Filter** (excellent for salt-and-pepper) or **Gaussian Filter**.

OpenCV Python Solution

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

print("Upload an image")
uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Read Image
img = cv2.imread(filename)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# -----
# Step 1 : Simulate Sensor Noise (Salt & Pepper)
# -----

noise_img = img.copy()

prob = 0.03
```

```

random = np.random.rand(*img.shape[:2])

noise_img[random < prob] = 0          # Black pixels
noise_img[random > 1-prob] = 255      # White pixels

# -----
# Step 2 : Remove Noise
# -----

filtered = cv2.medianBlur(noise_img,5)

# -----
# Display Images
# -----

plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(noise_img)
plt.title("Sensor Noisy Image")
plt.axis("off")

plt.subplot(1,3,3)
plt.imshow(filtered)
plt.title("Noise Removed")
plt.axis("off")

plt.show()

# -----
# Output
# -----

print("\n===== IMAGE ACQUISITION REPORT =====")

```

```
print("Input Image          : Successfully Loaded")
print("Sensor Noise Added   : Salt & Pepper Noise")
print("Noise Removal        : Median Filter")
print("Image Quality         : Improved")
print("Application           : Camera Sensor Processing")
print("Result                 : Noise Reduced Successfully")
print("=====")
```

Expected Output

The output shows a highly speckled, noisy image next to a perfectly restored, flat gray image. The median filter calculates out extreme outliers without blurring actual structure.